

VIJAYKANTH BONKURI

vbonkuri@wvstateu.edu | +1 (304) 419-8752 | LinkedIn | *Institute, WV, United States*

Research interests: Plant-microbe interactions · Rhizosphere biogeochemistry · Metagenomics and metatranscriptomics · Microbial community ecology · Microbiome bioinformatics · Machine learning

EDUCATION

West Virginia State University (WVSU)

Institute, WV

M.S. in Biotechnology (GPA: 4.0/4.0)

Aug 2025 – May 2027 (expected)

Thesis: Metatranscriptomic Analysis of Seasonal Temperature Effects on Methanogenesis in Kanawha River Sediments. **Advisor:** Dr. David Huber.

Relevant coursework: Biotechniques I & II, Biostatistics, Bioinformatics, Integrated Environmental Microbiology, Current Concepts in Biotechnology, Machine Learning & Data Mining, Analytical Chemistry.

Professor Jayashankar Telangana Agricultural University (PJTU)

Hyderabad, India

B.Sc. (Hons.) in Agricultural Sciences (GPA: 3.4/4.0)

Aug 2018 – Jul 2022

Agricultural Experiential Learning Project: Application of *Azotobacter* in Rice Cultivation for Biological Nitrogen Fixation and Plant Growth Promotion. **Advisor:** Dr. P. Arun Kumar.

Relevant coursework: Plant Pathology, Agricultural Microbiology, Soil Sciences, Statistics, Genetics & Plant Breeding, Biochemistry, Biotechnology, Agronomy, Agricultural Economics & Extension.

Professor Jayashankar Telangana Agricultural University (PJTU)

Hyderabad, India

Diploma in Agricultural Sciences (GPA: 4.0/4.0)

Jun 2016 – May 2018

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Institute, WV

Dr. David Huber's Environmental Microbiology & Biotechnology Lab, WVSU

Aug 2025 – Present

Projects

1. Evaluating the Impacts of Various Contaminants on Sediment Microbiomes and Methane Emission in the Kanawha River Ecosystem

- Examining how salinity, petroleum-derived contaminants, and pesticides affect sediment microbial communities and methane production. The study combined anaerobic sediment microcosms, gas chromatography, and 16S rRNA gene community analysis to assess contaminant-associated changes in methane production potential and microbial composition.

2. Assessing the Microbial Ecology and Geochemistry of Engineered Passive Remediation of an Acid Mine Drainage Stream in Central Appalachia

- Contributed to a study assessing passive acid mine drainage remediation in Cabin Creek. The study integrated water and sediment geochemistry with 16S rRNA gene community analysis to compare treatment ponds and stream locations, examine microbial-environmental relationships, and evaluate changes relative to pre-remediation conditions.

Young Professional – I (Research Position)

Nagpur, India

Central Institute for Cotton Research (CICR), ICAR

Jul 2024 – May 2025

- Contributed to the project “Targeting technologies to agro-ecological zones: large-scale demonstrations of best practices to enhance cotton productivity.”
- Applied biofertilizer treatments, measured soil microbial biomass, and performed microbial isolation and enumeration in cotton field demonstrations under high-density planting systems; compiled data for regional reporting.

Research Intern

Kethari Agri Tech Private Limited

Telangana, India

Sep 2022 – Mar 2023

- Evaluated applications of the biofertilizer *Pseudomonas fluorescens* combined with chemical fertilizers for biotic and abiotic stress management in maize and cotton cropping systems.
- Prepared *Pseudomonas fluorescens* inoculants and conducted cotton and maize field trials with microbial and fertilizer treatments; quantified growth, yield, and disease responses and analyzed treatment effects.

Research Assistant

Regional Agricultural Research Station (RARS), PJTAU

Hyderabad, India

Aug 2021 – Nov 2021

- Assisted with long-term fertilizer trials examining the effects of nutrient regimes on soil microbial ecology and microbial biomass under AICRP on Rice.

COMPUTATIONAL RESEARCH – RESEARCH TRAINEE

K-Means Clustering and Compositional Data Analysis of Cotton Rhizosphere Microbial Communities

Feb – Aug 2026

BioTecNika AI/ML in Biology and Bioinformatics Training Program

Bangalore, India

Mentor: Dr. Bhupender Singh

- **Machine learning training:** Supervised and unsupervised learning (K-nearest neighbors, decision trees, random forests, regression, K-means, and hierarchical clustering); data preprocessing, feature scaling, and model evaluation; introductory neural networks and ensemble methods.
- **Research project:** Applied K-means clustering following centered log-ratio (CLR) transformation to bacterial 16S rRNA gene and fungal ITS amplicon data from a published cotton rhizosphere fungicide seed-coating study to characterize community structure across geographic and treatment gradients.
- Evaluated cluster robustness and interpreted cluster-associated differences in predicted functional pathways, differential taxon abundance, and ecological guilds and enzymatic traits to explore potential relationships with fungicide degradation and microbial community function.

TEACHING EXPERIENCE

Graduate Teaching Assistant – WVSU

Aug 2025 – May 2026

Course: BIOL 341 – Microbiology

- Prepared experimental materials and equipment, managed laboratory setup and cleanup, and supported undergraduate microbiology instruction under Dr. David Huber.
- Assisted and supervised students during microbiology experiments, supported laboratory safety practices, and graded examinations.

Teaching Assistant – PJTAU

Feb 2020 – Jul 2020

Course: AGRO-101 – Fundamentals of Agronomy and Soil Science

- Assisted diploma students with soil-testing laboratory experiments, coordinated laboratory sessions and field practicals, and graded laboratory worksheets.

TECHNICAL SKILLS

- **Bioinformatics & Statistics:** QIIME 2, Cutadapt, DADA2, SILVA, UNITE, BLAST, MEGA, PICRUSt2; R (phyloseq, vegan, ggplot2, ALDEx2); Python (NumPy, pandas, scikit-learn, Matplotlib). Experience with microbial diversity analysis, centered log-ratio (CLR) transformation, PERMANOVA, PCA, K-means clustering, and cluster stability assessment.
- **Metagenomics & Metatranscriptomics:** Assembly-based sequencing workflows, rRNA removal, gene prediction, functional annotation, and transcript quantification, and differential expression analysis; experience with MEGAHIT, DIAMOND, Salmon, and DESeq2. Familiarity with hybrid assembly and metagenome-assembled genome (MAG) recovery strategies.

- **Molecular Biology:** Environmental nucleic acid extraction, PCR, RT-qPCR, agarose gel electrophoresis, cDNA synthesis, gene cloning, plasmid extraction, and competent cell preparation.
- **Microbiology:** Microbial isolation, cultivation, enumeration, characterization, and identification; media preparation, serial dilution, microscopy, differential staining (Gram, capsule, acid-fast, and endospore), biofertilizer inoculant preparation, and biocontrol-agent production.
- **Instrumentation:** GC for greenhouse gas measurement; YSI; UV–Vis spectrophotometer; Systronics flame photometer; Kjeldahl distillation apparatus; Qubit and NanoDrop; tensiometer and soil thermometer; microcentrifuges; incubators and Microscopy.
- **Environmental & Agricultural Methods:** Soil and sediment sampling and homogenization, nutrient analysis, moisture profiling, field trial design, and assessment of crop growth, yield, and disease responses.
- **Scientific Computing:** Linux/Bash, WSL2, conda, Jupyter notebooks, Git/GitHub, reproducible data analysis, and introductory HPC experience with SLURM.

PUBLICATIONS

- K-means clustering and compositional data analysis of cotton rhizosphere microbial communities. (In preparation)
1. B Raja MadhuShekar, B Anil Kumar, K Mamatha, G Narender, M Malla Reddy, M Venu, K Shankar Ganesh, R Roopa, and **B Vijaykanth**. (2025). Impact of high density planting system demonstrations on cotton farmers in Bhuvanagiri and Jangaon districts of Telangana. *International Journal of Agriculture Extension and Social Development*, 8(Special Issue 1), 113–118. doi:10.33545/26180723.2025.v8.i1Sb.1588.
 2. B Anil Kumar, M Venu, B Raja Madhushekar, K Mamatha, and **B Vijaykanth**. (2024). Adoption level of high density planting system in cotton crop at Jangaon district of Telangana state. *International Journal of Research in Agronomy*, SP-7(12), 345–348. doi:10.33545/2618060X.2024.v7.i12Sf.2174.
 3. **Vijaykanth Bonkuri**. (2024). A Review on Modern Agronomic Practices. *International Journal of Advanced Research in Science and Technology*, 13(4), 1339–1343. doi:10.62226/ijarst20241302.

PRESENTATIONS

- Oral presentation on: “Assessing the Microbial Ecology and Geochemistry of Engineered Passive Remediation of an Acid Mine Drainage Stream in Central Appalachia.” Graduate Research Symposium, WVSU, 2026.
- Poster presentation: “Application of *Azotobacter* in Rice Cultivation for Biological Nitrogen Fixation and Plant Growth Promotion.” Student Research Expo organized by ICAR and PJTSAU, 2022.
- Oral presentation: “Biochar for Sustainable Agriculture.” Cluster-level competition, Mahatma Gandhi National Council of Rural Education (MGNCRE), World Soil Day, Dec 5, 2020.

CONFERENCES AND WORKSHOPS

- DST-funded ICRISAT Workshop on Regenerative Agriculture, India, Sep 17, 2022.
- Workshop on Climate Change, Agriliclinics and Agribusiness Centers, NABARD & PJTAU, Dec 22, 2021.
- National Seminar: Entrepreneurship Awareness Camp, Department of Extension, PJTSAU.
- Attended an international training-cum-workshop program on “Agriculture 5.0: Next-Level Approaches to Smart Farming, Innovation, and Agripreneurship,” organized by ICAR and AEEFWS.
- Short Course: Representing Agriculture through Digital Comics, Ag Hub, PJTSAU.

LEADERSHIP, SERVICE, AND PROFESSIONAL MEMBERSHIPS

- **Member** – American Society for Microbiology (ASM).
- **Member** – International Society for Microbial Ecology (ISME).
- **Student Coordinator** – All India Agricultural Students Association (AIASA), 2019–2022.
- **Team Leader & Organizer** – Agricultural Experiential Learning Program (AELP), six-month program.